

MAE5834 - Simulation Study on Robust Inference - Gamma

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1 Introduction

This work aims to evaluate the performance of the MDPDE in comparison to the MLE under different contamination scenarios and tuning constant values $\alpha \in [0, 1)$. We drew a number from 1 to 10 in R Core Team (2026) to define the model to be worked on. Model number 3 was drawn, therefore, we will work with the Gamma regression model. In the following sections, we will define the model, present the adopted methodologies, describe the scenarios, and discuss the results.

2 Model and Methodologies

Let Y be a random variable with a Gamma distribution with mean μ and coefficient of variation $\phi^{-\frac{1}{2}}$. The probability density function of Y is given by:

$$f_Y(y, \mu, \phi) = exp\left\{\phi\left[\frac{-y}{\mu} - log(\mu)\right] - log(\Gamma(\phi)) + \phi log(\phi y) - log(y)\right\},$$

where $y > 0$, $\phi > 0$, $\mu > 0$, and $\Gamma(\phi) = \int_0^\infty t^{\phi-1}e^{-t}dt$ is the gamma function.Let $Y_1, \cdots, Y_n$ be a random sample with

$$Y_i|\mathbf{x_i} \overset{\text{ind}}{\sim} Gamma(\mu_i, \phi)$$
$$\eta_i = log(\mu_i), \eta_i = \mathbf{x_i}^\top \boldsymbol{\beta}$$

In this work, we will keep ϕ fixed, thus, the parameter vector is $\boldsymbol{\beta} = (\beta_0, \beta_1)^\top$. The score vector can be obtained by differentiating the log-likelihood with respect to $\boldsymbol{\beta}$. For the i-th observation, it is given by:

$$u_{\boldsymbol{\beta}}(y_i) = \phi\frac{(y_i - \mu_i)}{\mu_i}\mathbf{x_i},$$

where $\mathbf{x_i}$ is the i-th row of matrix $\mathbf{X}$. For the full sample, it can be expressed as

$$U_{\boldsymbol{\beta}}(\mathbf{y}) = \phi\mathbf{X}^\top\mathbf{V}^{\frac{-1}{2}}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

where $\mathbf{V}^{-\frac{1}{2}} = \text{diag}\left\{\frac{1}{\mu_1}, \cdots, \frac{1}{\mu_n}\right\}$, since the variance function for the Gamma distribution is $V(\mu) = \mu^2$.

2.1 Maximum Likelihood Estimator (MLE)

The maximum likelihood estimates for $\boldsymbol{\beta}$ will be obtained using the *gamlss* package (Rigby and Stasinopoulos, 2005).

2.2 Minimum Density Power Divergence Estimator (MDPDE)

To obtain the MDPDE estimates, we constructed an estimation algorithm. The MDPDE minimizes the following loss function:

$$H_n(\boldsymbol{\beta}) = \int_{\mathcal{X}} f_Y^{1+\alpha}(z;\mu_i,\phi)dz - \left(1+\frac{1}{\alpha}\right)\frac{1}{n}\sum_{i=1}^{n} f_Y^{\alpha}(Y_i;\mu_i,\phi), \text{ for } \alpha \in [0,1).$$

As $\alpha \to 0$ 0, the MDPDE reduces to the MLE. The MDPDE belongs to the class of M-estimators. It satisfies estimating equations defined by the score-weighted ψ-function:

$$\psi(y_i,\beta) = u_\beta(y_i) f_Y^{\alpha}(y_i;\mu_i,\phi) - \int_{\mathcal{X}} u_\beta(z) f_Y^{1+\alpha}(z;\mu,\phi)(z)dz$$

And the estimates are obtained by solving the system of estimating equations:

$$\sum_{i=1}^{n} \psi_i(y_i,\beta) = \mathbf{0}$$

2.3 Estimation Algorithm: Iteratively Reweighted Least Squares (IRLS / Fisher Scoring) for MDPDE

Following the classical Generalized Linear Model (GLM) framework and directly reflecting our R implementation, the iterative procedure for obtaining the Minimum Density Power Divergence Estimator (MDPDE) in Gamma regression is structured as follows:

2.3.1 Step 1: Initialization ($m = 0$)

The initial parameter vector $\boldsymbol{\beta}^{(0)}$ is initialized with the Maximum Likelihood Estimates (MLE) fitted via the `gamlss` package:

$$\boldsymbol{\beta}^{(0)} = \hat{\boldsymbol{\beta}}_{\text{MLE}}$$

2.3.2 Step 2: Updating Quantities at Iteration m

Given the current parameter estimates $\boldsymbol{\beta}^{(m)}$ at iteration m:

1. **Linear Predictor and Mean Vector ($\boldsymbol{\mu}$):**

$$\boldsymbol{\eta}^{(m)} = \mathbf{X}\boldsymbol{\beta}^{(m)}, \quad \mu_i^{(m)} = \exp\left(\eta_i^{(m)}\right), \quad i = 1, \ldots, n$$

2. **Robust Weight Matrix (W):** Calculate the density power weights $w_i^{(m)}$ for each observation using the Gamma probability density function $f_Y(y_i; \mu_i^{(m)}, \phi)$:

$$w_i^{(m)} = \left[f_Y(y_i; \mu_i^{(m)}, \phi)\right]^{\alpha}, \quad \mathbf{W}^{(m)} = \operatorname{diag}\left\{w_1^{(m)}, \ldots, w_n^{(m)}\right\}$$

3. **Adjusted Response Vector ($\mathbf{z}_{\mathbf{adj}}$):** Form the working response vector based on the log-link GLM linearization:

$$\mathbf{z}_{\text{adj}}^{(m)} = \boldsymbol{\eta}^{(m)} + \frac{\mathbf{y} - \boldsymbol{\mu}^{(m)}}{\boldsymbol{\mu}^{(m)}}$$

4. **Integral Bias Correction Term ($\mathbf{b}_{\mathbf{soma}}$):** Evaluate the integral penalty vector $\mathbf{b}_{\text{soma}}^{(m)}$ required for the MDPDE score equations.

2.3.3 Step 3: Matrix Computations and Parameter Update

Compute the weighted Hessian matrix and the working response projection vector:

$$\mathbf{H}^{(m)} = \phi \mathbf{X}^{\top} \mathbf{W}^{(m)} \mathbf{X}$$

$$\mathbf{V}_z^{(m)} = \phi \mathbf{X}^{\top} \mathbf{W}^{(m)} \mathbf{z}_{\text{adj}}^{(m)}$$

The parameter update at step $m+1$ is obtained by solving the linear system:

$$\boldsymbol{\beta}^{(m+1)} = \left(\mathbf{H}^{(m)}\right)^{-1} \left(\mathbf{V}_z^{(m)} - \mathbf{b}_{\text{soma}}^{(m)}\right)$$

2.3.3.1 Closed-Form Solution of the Integral Bias Vector ($\mathbf{b}_{\mathbf{soma}}$)

To ensure Fisher consistency for the MDPDE when $\alpha > 0$, the integral bias correction term is computed analytically:

$$\mathbf{b}_{\text{soma}} = \mathbf{X}^{\top} \mathbf{v}_b$$

where $\mathbf{v}_b = (b_{1,\text{scalar}}, \ldots, b_{n,\text{scalar}})^{\top}$ is an $n \times 1$ vector whose elements are defined as:

$$b_{i,\text{scalar}} = \frac{\phi^{1-\phi\alpha} \cdot \alpha \cdot \Gamma\big((1+\alpha)\phi - \alpha\big)}{\mu_i^{\alpha} \cdot (1+\alpha)^{(1+\alpha)\phi+1} \cdot \Gamma^{1+\alpha}(\phi)}$$

2.3.4 Step 4: Convergence Check

Compute the sum of squared differences between successive parameter vectors:

$$\text{criterio} = \sum_{j=0}^{p} \left(\beta_j^{(m+1)} - \beta_j^{(m)}\right)^2$$

- **If criterio** $< \delta$**:** The algorithm stops and returns the converged estimates $\hat{\boldsymbol{\beta}}_{\text{MDPDE}} = \boldsymbol{\beta}^{(m+1)}$.
- **Otherwise:** Set $\boldsymbol{\beta}^{(m)} \leftarrow \boldsymbol{\beta}^{(m+1)}$ and return to **Step 2**.

3 Simulation Study

The true model, that is, the one that will generate the samples, is defined as follows:

$$Y_i|\mathbf{x_i} \overset{\text{ind}}{\sim} Gamma(\mu_i, \phi)$$
$$\eta_i = log(\mu_i), \eta_i = \beta_0 + \beta_1 x_{i1}$$

where

$$\boldsymbol{\beta} = (0.3, 0.1)^\top;\ \phi = 4;\ x_{i1} \sim U(1, 10);\ i = 1, \cdots, n.$$

3.1 Scenario C0 (No contamination):

Pure data generated strictly by the theoretical model.

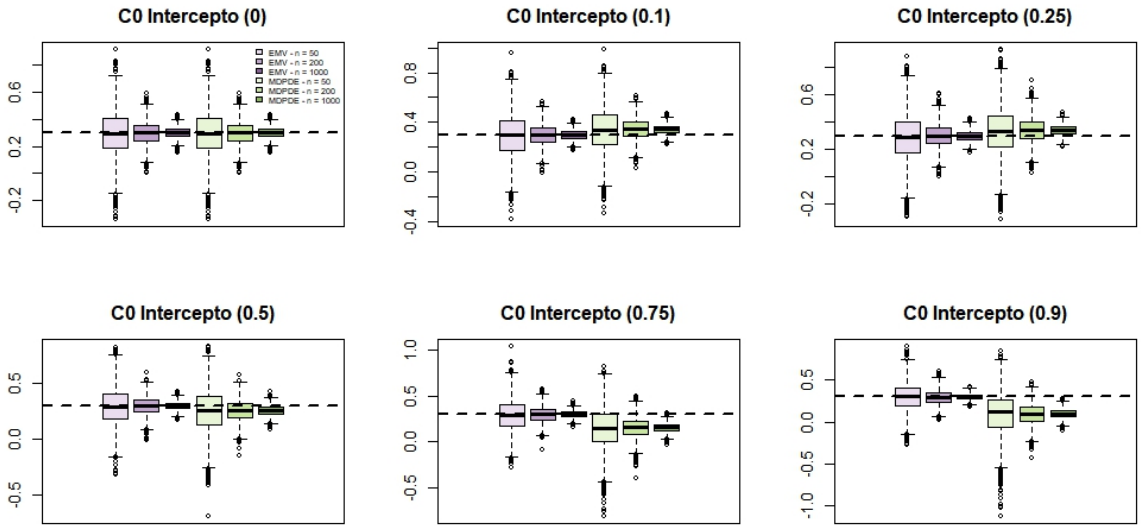


Figure 1: Distributions of intercept estimates for each sample size.

In Figure 1, six distributions are displayed in the same plot. The first three correspond to the Maximum Likelihood Estimator (MLE) for sample sizes $n = \{50, 200, 1000\}$, while the remaining three correspond to the Minimum Density Power Divergence Estimator (MDPDE). The values of the tuning parameter $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9\}$ are given in parentheses. As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution for the estimates.

Table 1: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-0.0070	0.0293	0.0292	-0.0070	0.0293	0.0292
200	-0.0029	0.0067	0.0067	-0.0029	0.0067	0.0067
1000	-0.0001	0.0014	0.0014	-0.0001	0.0014	0.0014
$\alpha = 0.1$						
50	-0.0073	0.0291	0.0291	0.0383	0.0311	0.0296
200	-0.0026	0.0070	0.0070	0.0430	0.0090	0.0071
1000	-0.0009	0.0013	0.0013	0.0445	0.0034	0.0014
$\alpha = 0.25$						
50	-0.0099	0.0292	0.0291	0.0308	0.0315	0.0305
200	0.0003	0.0069	0.0069	0.0392	0.0089	0.0074
1000	-0.0006	0.0014	0.0014	0.0382	0.0030	0.0015
$\alpha = 0.5$						
50	-0.0056	0.0285	0.0284	-0.0458	0.0401	0.0380
200	0.0003	0.0068	0.0069	-0.0443	0.0107	0.0087
1000	-0.0002	0.0015	0.0015	-0.0428	0.0037	0.0019
$\alpha = 0.75$						
50	-0.0096	0.0289	0.0288	-0.1564	0.0742	0.0497
200	-0.0008	0.0071	0.0071	-0.1436	0.0328	0.0122
1000	-0.0007	0.0014	0.0014	-0.1438	0.0230	0.0023
$\alpha = 0.9$						
50	-0.0019	0.0276	0.0276	-0.2084	0.1079	0.0645
200	-0.0043	0.0072	0.0072	-0.2097	0.0591	0.0151
1000	-0.0004	0.0013	0.0013	-0.2037	0.0444	0.0029

However, as α approaches 1, the MLE exhibits superior performance, whereas the MDPDE displays bias.

In Table 1, we observe that the MLE presents better results in terms of MSE. For $\alpha > 0$, the MLE becomes more competitive, as the MDPDE is known to suffer from a loss of efficiency.

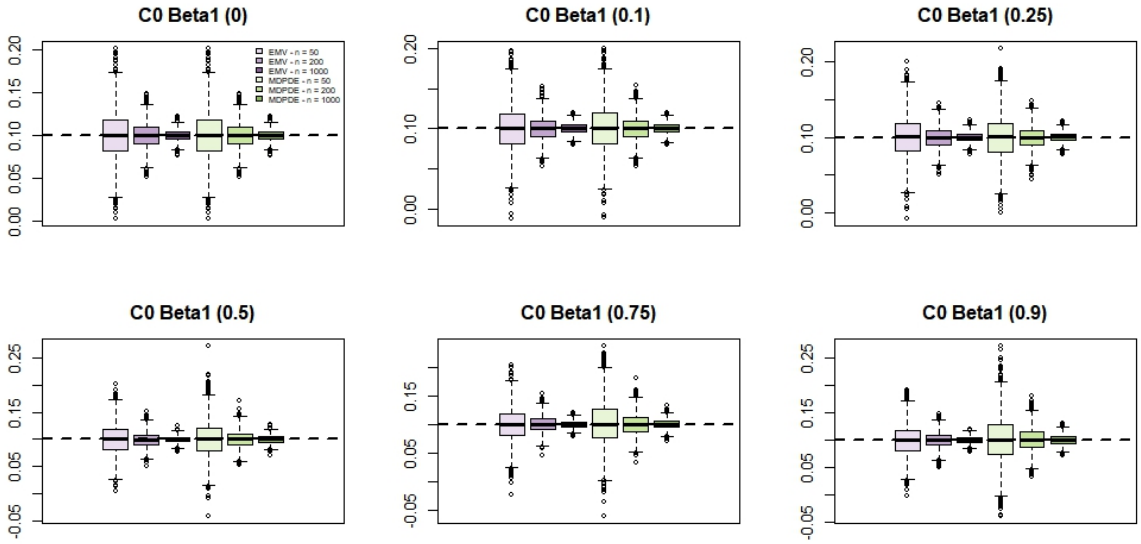


Figure 2: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 2: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	4e-04	8e-04	8e-04	0.0004	0.0008	0.0008
200	2e-04	2e-04	2e-04	0.0002	0.0002	0.0002
1000	0e+00	0e+00	0e+00	0.0000	0.0000	0.0000
$\alpha = 0.1$						
50	2e-04	8e-04	8e-04	0.0003	0.0008	0.0008
200	1e-04	2e-04	2e-04	0.0001	0.0002	0.0002
1000	1e-04	0e+00	0e+00	0.0001	0.0000	0.0000
$\alpha = 0.25$						
50	5e-04	8e-04	8e-04	0.0004	0.0008	0.0008
200	-2e-04	2e-04	2e-04	-0.0002	0.0002	0.0002
1000	0e+00	0e+00	0e+00	0.0000	0.0000	0.0000
$\alpha = 0.5$						
50	3e-04	8e-04	8e-04	0.0003	0.0010	0.0010
200	-2e-04	2e-04	2e-04	0.0003	0.0002	0.0002
1000	0e+00	0e+00	0e+00	0.0000	0.0000	0.0000
$\alpha = 0.75$						
50	2e-04	8e-04	8e-04	0.0015	0.0014	0.0014
200	-1e-04	2e-04	2e-04	0.0003	0.0003	0.0003
1000	1e-04	0e+00	0e+00	0.0003	0.0001	0.0001
$\alpha = 0.9$						
50	-5e-04	8e-04	8e-04	0.0009	0.0017	0.0017
200	3e-04	2e-04	2e-04	0.0012	0.0004	0.0004
1000	1e-04	0e+00	0e+00	0.0003	0.0001	0.0001

For β_1, Figure 2 and Table 2 show that the MLE achieves superior performance.

In the scenario without contamination (pure data), the MDPDE suffers in the trade-off against the Maximum Likelihood Estimator (MLE). Because the data strictly follow the assumed model, the MLE is the most efficient estimator, yielding lower MSE values.

3.2 Scenario C1 (5% of contamination):

In this scenario, 5% of the observations had their response values replaced by a value generated from a Gamma distribution with a mean 5 times the original mean value of the observation

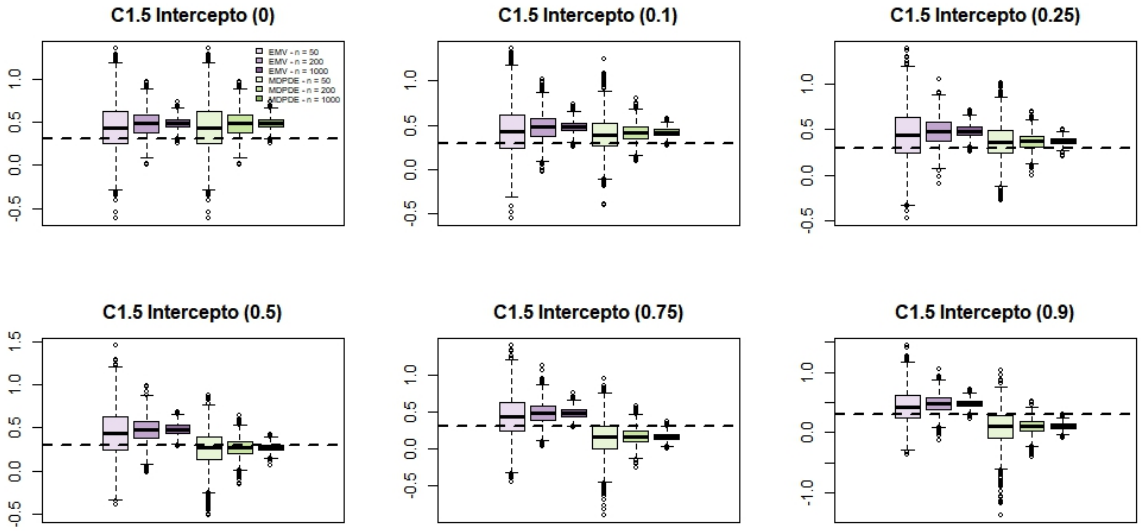


Figure 3: Distributions of intercept estimates for each sample size.

In Figure 3, six distributions are displayed in the same plot. The first three correspond to the Maximum Likelihood Estimator (MLE) for sample sizes $n = \{50, 200, 1000\}$, while the remaining three correspond to the Minimum Density Power Divergence Estimator (MDPDE). The values of the tuning parameter $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9\}$ are given in parentheses. As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution for the estimates. For $\alpha = 0.5$ MDPDE appear to have superior performance.

In Table 3, we observe that for $\alpha \in [0.1, 0.75]$, the MDPDE achieves a lower MSE. Consequently, it is preferred under sample contamination. For higher tuning parameters, such as $\alpha = 0.9$, the variance-bias trade-off favors the MLE.

Table 3: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	0.1389	0.0966	0.0773	0.1389	0.0966	0.0773
200	0.1811	0.0544	0.0217	0.1811	0.0544	0.0217
1000	0.1819	0.0374	0.0044	0.1819	0.0374	0.0044
$\alpha = 0.1$						
50	0.1328	0.0957	0.0781	0.0968	0.0488	0.0394
200	0.1784	0.0537	0.0219	0.1192	0.0240	0.0099
1000	0.1813	0.0375	0.0046	0.1196	0.0164	0.0021
$\alpha = 0.25$						
50	0.1408	0.0965	0.0767	0.0629	0.0393	0.0353
200	0.1770	0.0530	0.0217	0.0681	0.0131	0.0084
1000	0.1825	0.0375	0.0042	0.0707	0.0067	0.0017
$\alpha = 0.5$						
50	0.1365	0.0969	0.0783	-0.0389	0.0403	0.0388
200	0.1744	0.0522	0.0218	-0.0303	0.0107	0.0098
1000	0.1840	0.0381	0.0042	-0.0297	0.0029	0.0020
$\alpha = 0.75$						
50	0.1393	0.0984	0.0790	-0.1510	0.0774	0.0546
200	0.1835	0.0552	0.0215	-0.1359	0.0309	0.0124
1000	0.1829	0.0379	0.0045	-0.1366	0.0212	0.0025
$\alpha = 0.9$						
50	0.1366	0.0937	0.0751	-0.2201	0.1217	0.0732
200	0.1803	0.0548	0.0223	-0.2005	0.0556	0.0154
1000	0.1808	0.0371	0.0044	-0.1978	0.0422	0.0030

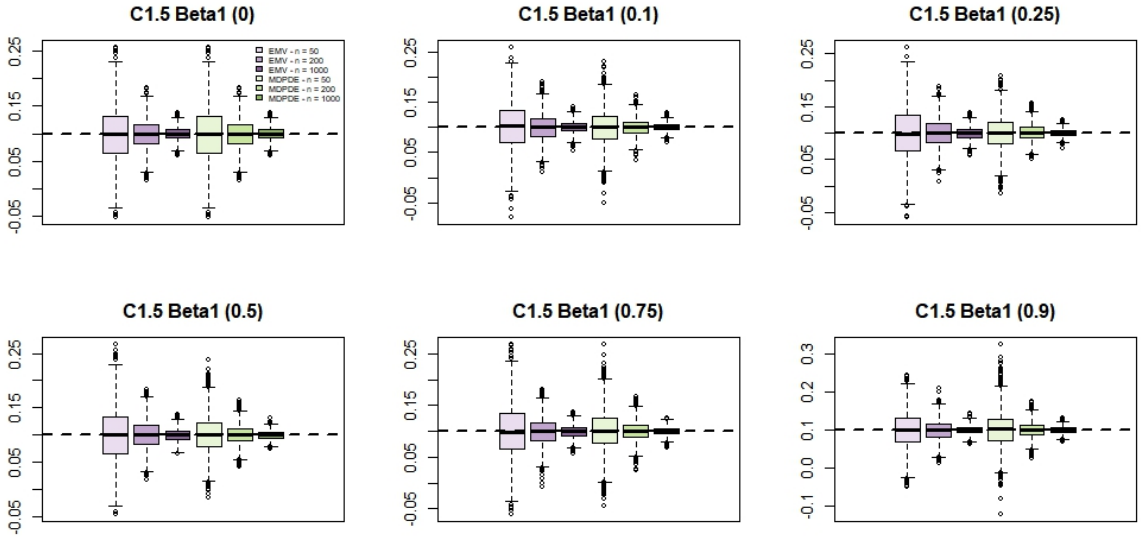


Figure 4: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 4: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-5e-04	0.0023	0.0023	-0.0005	0.0023	0.0023
200	1e-04	0.0006	0.0006	0.0001	0.0006	0.0006
1000	0e+00	0.0001	0.0001	0.0000	0.0001	0.0001
$\alpha = 0.1$						
50	5e-04	0.0023	0.0023	0.0005	0.0011	0.0011
200	1e-04	0.0007	0.0007	0.0000	0.0003	0.0003
1000	1e-04	0.0001	0.0001	0.0001	0.0001	0.0001
$\alpha = 0.25$						
50	-6e-04	0.0023	0.0023	-0.0002	0.0010	0.0010
200	1e-04	0.0007	0.0007	0.0003	0.0002	0.0002
1000	-2e-04	0.0001	0.0001	0.0001	0.0000	0.0000
$\alpha = 0.5$						
50	6e-04	0.0023	0.0023	0.0012	0.0011	0.0011
200	8e-04	0.0007	0.0007	0.0002	0.0003	0.0003
1000	-3e-04	0.0001	0.0001	0.0000	0.0001	0.0001
$\alpha = 0.75$						
50	-6e-04	0.0024	0.0024	0.0017	0.0014	0.0014
200	-9e-04	0.0007	0.0007	0.0001	0.0003	0.0003
1000	-2e-04	0.0001	0.0001	0.0003	0.0001	0.0001
$\alpha = 0.9$						
50	2e-04	0.0022	0.0022	0.0030	0.0020	0.0020
200	-4e-04	0.0007	0.0007	0.0006	0.0004	0.0004
1000	1e-04	0.0001	0.0001	0.0002	0.0001	0.0001

For β_1, Figure 4 presents no indication that the MDPDE is preferred. However, Table 4 demonstrates that the MDPDE achieves superior performance. In the scenario with 5% contamination, the trade-off favors the Maximum Likelihood Estimator (MLE) over the MDPDE, as the light contamination level is insufficient to offset the efficiency loss associated with robust estimation.

3.3 Scenario C1 (10% of contamination):

In this scenario, 10% of the observations had their response values replaced by a value generated from a Gamma distribution with a mean 5 times the original mean value of the observation

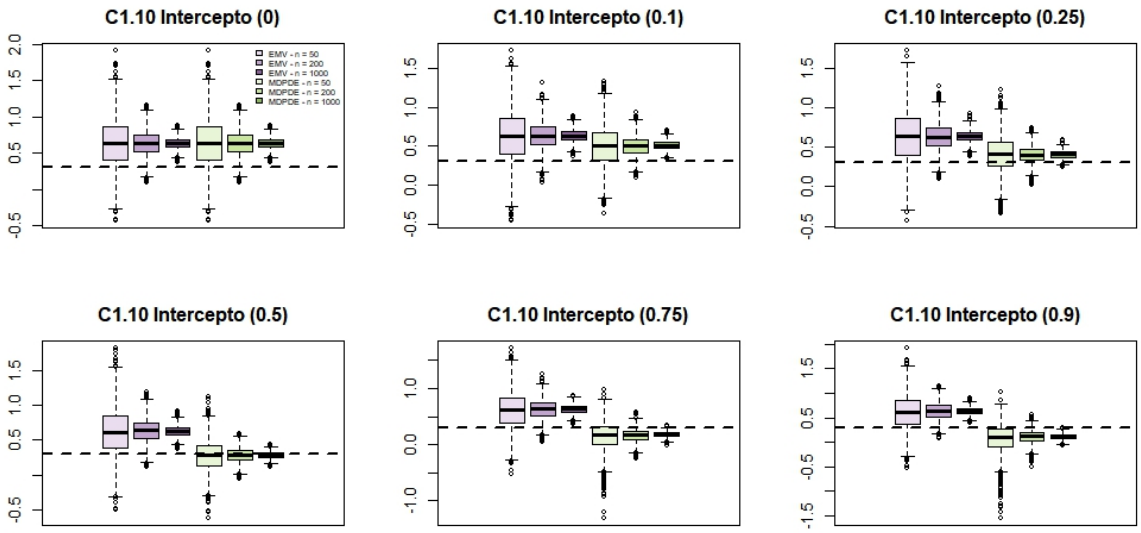


Figure 5: Distributions of intercept estimates for each sample size.

In Figure 5, six distributions are displayed in the same plot. The first three correspond to the Maximum Likelihood Estimator (MLE) for sample sizes $n = \{50, 200, 1000\}$, while the remaining three correspond to the Minimum Density Power Divergence Estimator (MDPDE). The values of the tuning parameter $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9\}$ are given in parentheses. As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution for the estimates. However, as α approaches 0.5, the MDPDE appear to have superior performance, whereas the MLE displays bias.

In Table 5, we observe that the MDPDE is preferred under sample contamination, achieving the lowest MSE at $\alpha = 0.5$.

Table 5: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	0.3347	0.2237	0.1117	0.3347	0.2237	0.1117
200	0.3333	0.1396	0.0285	0.3333	0.1396	0.0285
1000	0.3388	0.1205	0.0057	0.3388	0.1205	0.0057
$\alpha = 0.1$						
50	0.3292	0.2215	0.1131	0.2050	0.1064	0.0644
200	0.3342	0.1408	0.0291	0.2066	0.0583	0.0156
1000	0.3347	0.1176	0.0056	0.2068	0.0458	0.0031
$\alpha = 0.25$						
50	0.3289	0.2226	0.1145	0.1107	0.0584	0.0461
200	0.3287	0.1364	0.0284	0.1047	0.0218	0.0109
1000	0.3363	0.1187	0.0056	0.1067	0.0136	0.0022
$\alpha = 0.5$						
50	0.3188	0.2138	0.1122	-0.0220	0.0461	0.0456
200	0.3350	0.1405	0.0283	-0.0164	0.0109	0.0106
1000	0.3354	0.1181	0.0056	-0.0152	0.0024	0.0022
$\alpha = 0.75$						
50	0.3141	0.2075	0.1089	-0.1391	0.0812	0.0619
200	0.3314	0.1383	0.0285	-0.1342	0.0317	0.0137
1000	0.3373	0.1195	0.0058	-0.1261	0.0186	0.0027
$\alpha = 0.9$						
50	0.3155	0.2143	0.1148	-0.2169	0.1261	0.0791
200	0.3346	0.1406	0.0286	-0.1917	0.0538	0.0171
1000	0.3357	0.1183	0.0057	-0.1907	0.0396	0.0032

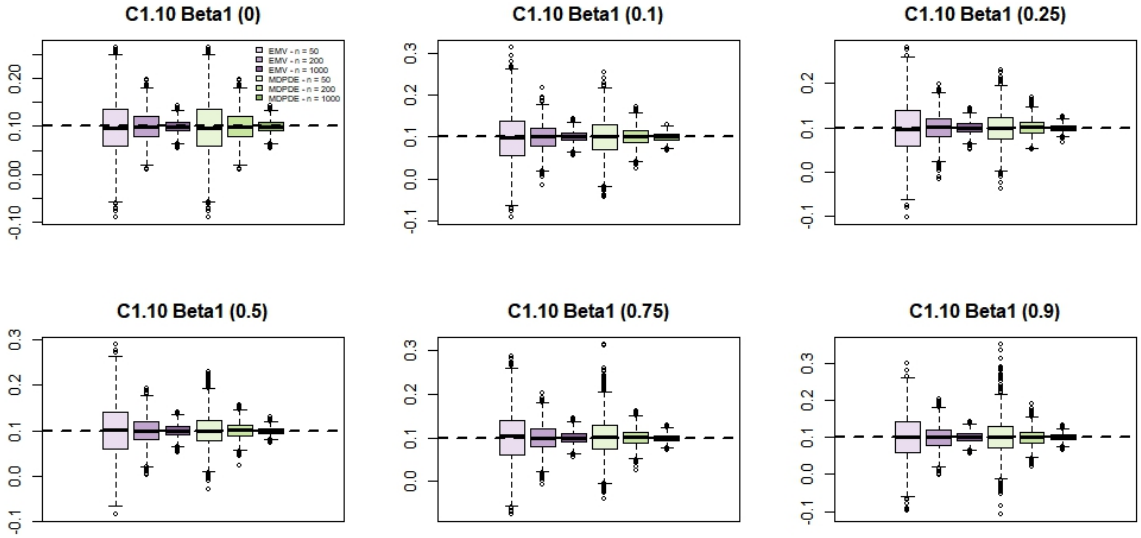


Figure 6: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 6: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-0.0027	0.0034	0.0034	-0.0027	0.0034	0.0034
200	-0.0001	0.0009	0.0009	-0.0001	0.0009	0.0009
1000	-0.0006	0.0002	0.0002	-0.0006	0.0002	0.0002
$\alpha = 0.1$						
50	-0.0011	0.0035	0.0035	0.0002	0.0019	0.0019
200	-0.0004	0.0009	0.0009	0.0003	0.0005	0.0005
1000	0.0001	0.0002	0.0002	0.0001	0.0001	0.0001
$\alpha = 0.25$						
50	-0.0009	0.0035	0.0035	-0.0007	0.0013	0.0013
200	0.0006	0.0009	0.0009	0.0004	0.0003	0.0003
1000	-0.0001	0.0002	0.0002	0.0001	0.0001	0.0001
$\alpha = 0.5$						
50	0.0001	0.0034	0.0034	0.0007	0.0012	0.0012
200	-0.0005	0.0009	0.0009	0.0002	0.0003	0.0003
1000	0.0000	0.0002	0.0002	-0.0001	0.0001	0.0001
$\alpha = 0.75$						
50	0.0010	0.0033	0.0033	0.0017	0.0017	0.0017
200	0.0001	0.0009	0.0009	0.0010	0.0004	0.0004
1000	-0.0003	0.0002	0.0002	-0.0001	0.0001	0.0001
$\alpha = 0.9$						
50	0.0012	0.0034	0.0034	0.0033	0.0021	0.0021
200	-0.0006	0.0009	0.0009	-0.0001	0.0005	0.0005
1000	0.0001	0.0002	0.0002	0.0000	0.0001	0.0001

For β_1, Figure 6 shows no clear preference for the MDPDE. However, Table 6 demonstrates that the MLE achieves superior performance.

In the scenario with 10% contamination, the trade-off strongly favors the MDPDE over the MLE, as the contamination degrades the performance of the Maximum Likelihood Estimator.

3.4 Scenario C2 (5% of contamination):

In this scenario, 5% of the observations had their respective values for the covariate x_{i1} generated from a continuous uniform distribution on the interval $(16, 17)$.

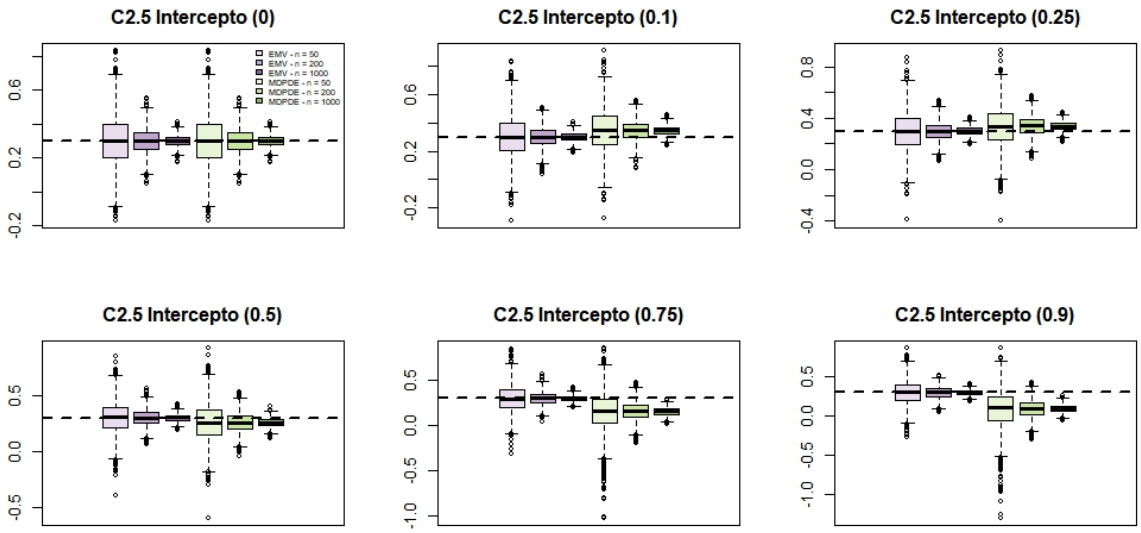


Figure 7: Distributions of intercept estimates for each sample size.

As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution for the estimates. However, as α approaches 1, the MLE appear to have superior performance, whereas the MDPDE displays bias.

In Table 7, we observe that MLE presents an better peformance for estimating the intercept in this contamination cenario.

Table 7: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-0.0025	0.0215	0.0215	-0.0025	0.0215	0.0215
200	-0.0006	0.0052	0.0052	-0.0006	0.0052	0.0052
1000	0.0002	0.0010	0.0010	0.0002	0.0010	0.0010
$\alpha = 0.1$						
50	0.0006	0.0217	0.0217	0.0455	0.0242	0.0222
200	0.0001	0.0048	0.0048	0.0450	0.0070	0.0049
1000	-0.0001	0.0010	0.0010	0.0452	0.0031	0.0010
$\alpha = 0.25$						
50	-0.0025	0.0225	0.0225	0.0337	0.0261	0.0249
200	0.0001	0.0050	0.0050	0.0390	0.0069	0.0054
1000	-0.0012	0.0010	0.0010	0.0374	0.0025	0.0011
$\alpha = 0.5$						
50	0.0016	0.0209	0.0209	-0.0448	0.0298	0.0278
200	0.0008	0.0048	0.0048	-0.0428	0.0082	0.0064
1000	0.0004	0.0010	0.0010	-0.0426	0.0032	0.0014
$\alpha = 0.75$						
50	-0.0040	0.0215	0.0215	-0.1486	0.0661	0.0440
200	0.0025	0.0050	0.0050	-0.1401	0.0293	0.0096
1000	-0.0010	0.0010	0.0010	-0.1434	0.0224	0.0019
$\alpha = 0.9$						
50	-0.0015	0.0222	0.0222	-0.2128	0.1045	0.0592
200	-0.0015	0.0051	0.0051	-0.2058	0.0543	0.0120
1000	0.0009	0.0010	0.0010	-0.2020	0.0431	0.0023

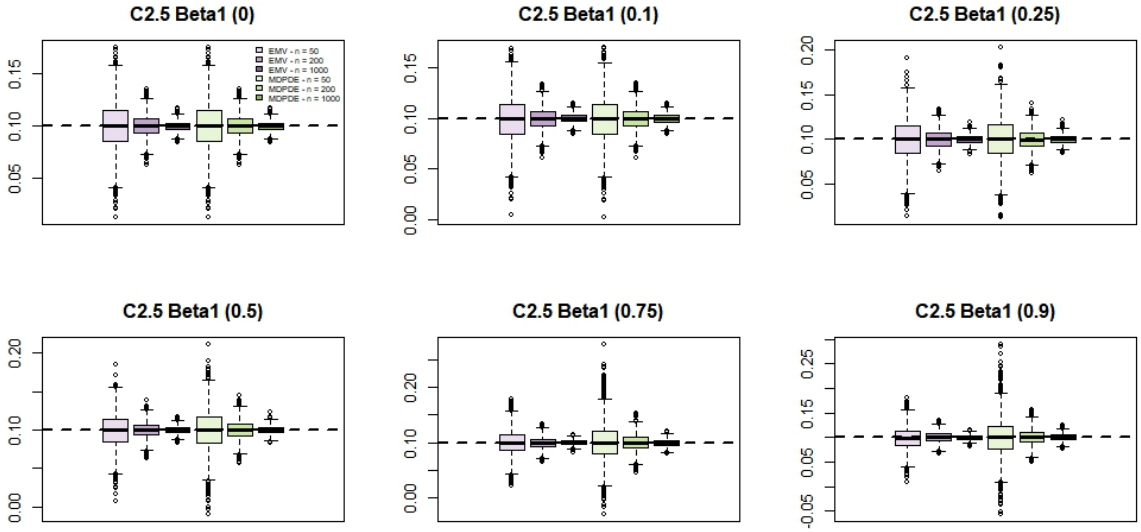


Figure 8: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 8: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-0.0004	5e-04	5e-04	-4e-04	0.0005	0.0005
200	0.0000	1e-04	1e-04	0e+00	0.0001	0.0001
1000	0.0000	0e+00	0e+00	0e+00	0.0000	0.0000
$\alpha = 0.1$						
50	-0.0009	5e-04	5e-04	-6e-04	0.0005	0.0005
200	-0.0002	1e-04	1e-04	-1e-04	0.0001	0.0001
1000	0.0000	0e+00	0e+00	0e+00	0.0000	0.0000
$\alpha = 0.25$						
50	-0.0004	5e-04	5e-04	3e-04	0.0005	0.0005
200	-0.0003	1e-04	1e-04	-3e-04	0.0001	0.0001
1000	0.0001	0e+00	0e+00	2e-04	0.0000	0.0000
$\alpha = 0.5$						
50	-0.0009	5e-04	5e-04	0e+00	0.0006	0.0006
200	-0.0004	1e-04	1e-04	0e+00	0.0001	0.0001
1000	-0.0001	0e+00	0e+00	0e+00	0.0000	0.0000
$\alpha = 0.75$						
50	-0.0003	5e-04	5e-04	4e-04	0.0010	0.0010
200	-0.0005	1e-04	1e-04	-1e-04	0.0002	0.0002
1000	0.0001	0e+00	0e+00	2e-04	0.0000	0.0000
$\alpha = 0.9$						
50	-0.0011	5e-04	5e-04	5e-04	0.0014	0.0014
200	0.0001	1e-04	1e-04	6e-04	0.0003	0.0003
1000	-0.0002	0e+00	0e+00	0e+00	0.0000	0.0000

For β_1, Figure 8 and Table 7 show that the MLE achieves superior performance.

In the scenario with 5% contamination, the MDPDE suffers in the trade-off against the Maximum Likelihood Estimator (MLE). The contamination was not severe enough to justify using a robust technique. Thus, the MLE remains the most efficient estimator, achieving minimum variance.

3.5 Scenario C2 (10% of contamination):

In this scenario, 10% of the observations had their respective values for the covariate x_{i1} generated from a continuous uniform distribution on the interval $(16, 17)$.

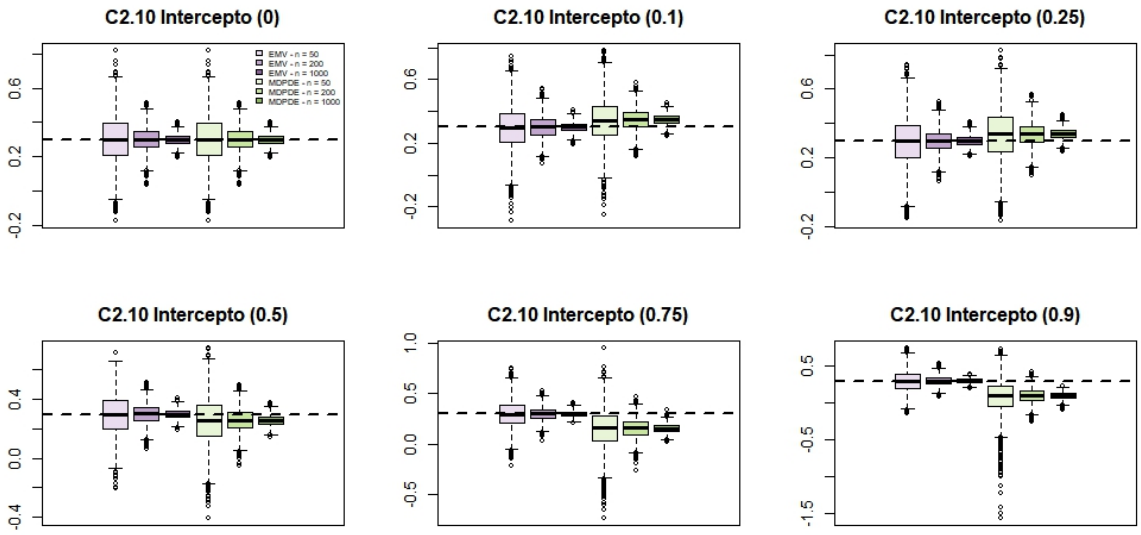


Figure 9: Distributions of intercept estimates for each sample size.

As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution of estimates. However, as α approaches 1, the MLE appears to achieve superior performance, whereas the MDPDE exhibits bias and greater dispersion.

In Table 9, we observe that the MLE achieves a lower MSE.

Table 9: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	0.0014	0.0183	0.0183	0.0014	0.0183	0.0183
200	0.0010	0.0046	0.0046	0.0010	0.0046	0.0046
1000	-0.0007	0.0009	0.0009	-0.0007	0.0009	0.0009
$\alpha = 0.1$						
50	-0.0044	0.0185	0.0185	0.0407	0.0203	0.0187
200	0.0008	0.0045	0.0045	0.0461	0.0067	0.0046
1000	0.0001	0.0009	0.0009	0.0455	0.0030	0.0009
$\alpha = 0.25$						
50	-0.0036	0.0192	0.0192	0.0341	0.0220	0.0208
200	-0.0006	0.0045	0.0045	0.0379	0.0064	0.0049
1000	-0.0005	0.0009	0.0009	0.0381	0.0024	0.0009
$\alpha = 0.5$						
50	-0.0046	0.0188	0.0188	-0.0502	0.0275	0.0250
200	-0.0001	0.0044	0.0044	-0.0440	0.0079	0.0059
1000	-0.0004	0.0009	0.0009	-0.0431	0.0031	0.0012
$\alpha = 0.75$						
50	-0.0013	0.0179	0.0179	-0.1470	0.0579	0.0363
200	-0.0011	0.0044	0.0044	-0.1442	0.0289	0.0081
1000	-0.0007	0.0009	0.0009	-0.1434	0.0223	0.0017
$\alpha = 0.9$						
50	0.0025	0.0193	0.0193	-0.2159	0.1002	0.0537
200	-0.0002	0.0042	0.0042	-0.2040	0.0521	0.0105
1000	0.0007	0.0009	0.0009	-0.2006	0.0424	0.0021

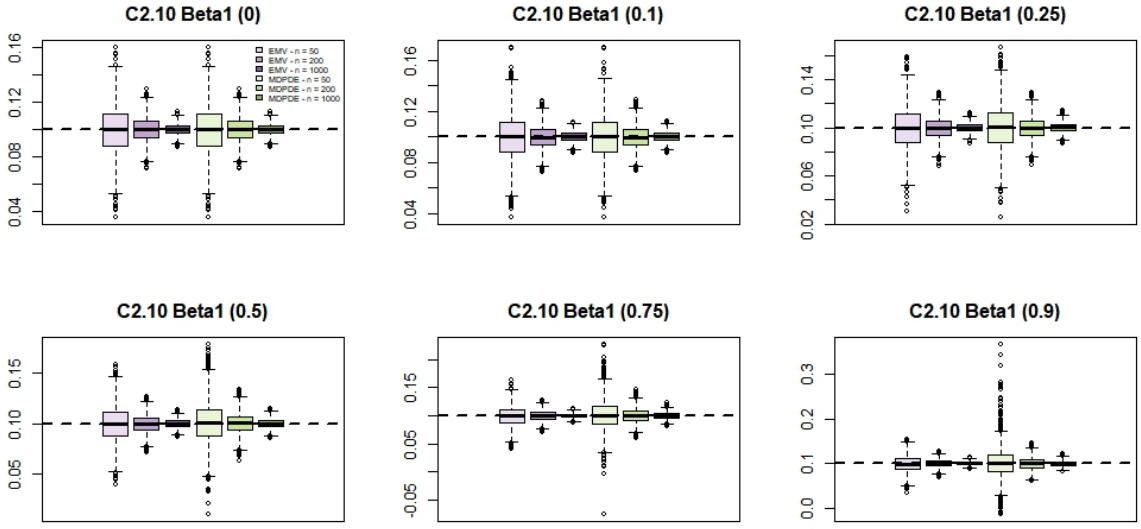


Figure 10: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 10: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-8e-04	3e-04	3e-04	-0.0008	3e-04	3e-04
200	-3e-04	1e-04	1e-04	-0.0003	1e-04	1e-04
1000	0e+00	0e+00	0e+00	0.0000	0e+00	0e+00
$\alpha = 0.1$						
50	-2e-04	3e-04	3e-04	0.0000	3e-04	3e-04
200	-3e-04	1e-04	1e-04	-0.0003	1e-04	1e-04
1000	0e+00	0e+00	0e+00	0.0000	0e+00	0e+00
$\alpha = 0.25$						
50	-5e-04	3e-04	3e-04	-0.0002	3e-04	3e-04
200	-2e-04	1e-04	1e-04	-0.0001	1e-04	1e-04
1000	0e+00	0e+00	0e+00	0.0000	0e+00	0e+00
$\alpha = 0.5$						
50	-3e-04	3e-04	3e-04	0.0006	4e-04	4e-04
200	-1e-04	1e-04	1e-04	0.0003	1e-04	1e-04
1000	0e+00	0e+00	0e+00	0.0001	0e+00	0e+00
$\alpha = 0.75$						
50	-3e-04	3e-04	3e-04	0.0005	7e-04	7e-04
200	0e+00	1e-04	1e-04	0.0002	1e-04	1e-04
1000	1e-04	0e+00	0e+00	0.0002	0e+00	0e+00
$\alpha = 0.9$						
50	-1e-03	3e-04	3e-04	0.0015	1e-03	1e-03
200	-3e-04	1e-04	1e-04	0.0001	2e-04	2e-04
1000	-1e-04	0e+00	0e+00	-0.0002	0e+00	0e+00

In Figure 10 and Table 10, we observe that the MLE achieves a lower or equivalent MSE. Consequently, it is preferred under the specific contamination pattern present in this scenario.

Even with 10% contamination, the MDPDE suffers in the trade-off against the Maximum Likelihood Estimator (MLE). For this type of contamination, the impact was not severe enough to justify a robust approach. Thus, the MLE remains the most suitable choice by preserving higher efficiency and lower variance.

3.6 Scenario C3 (5% of contamination):

In this scenario, 5% of the observations simultaneously had their responses replaced by a value generated from a Gamma distribution with a mean 5 times the original value, and their respective values for the covariate x_{i1} generated from a continuous uniform distribution on the interval $(16, 17)$.

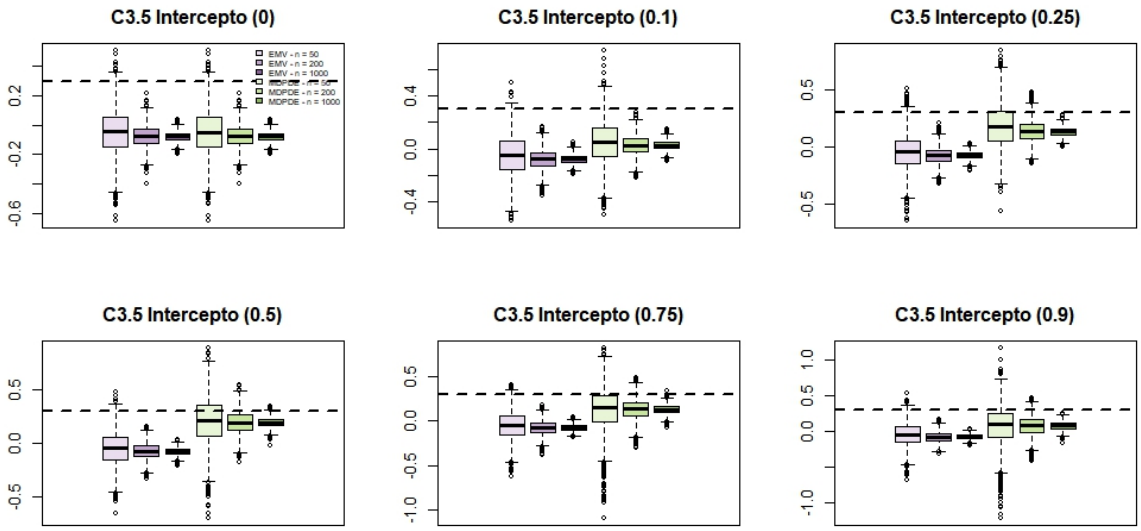


Figure 11: Distributions of intercept estimates for each sample size.

As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution for the estimates. For $\alpha = 0.5$ the MDPDE appear to have superior performance.

In Table 11, we confirm that for larger sample sizes, the MDPDE achieves better performance in terms of MSE at $\alpha = 0.5$.

Table 11: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-0.3504	0.1465	0.0237	-0.3504	0.1465	0.0237
200	-0.3788	0.1490	0.0055	-0.3788	0.1490	0.0055
1000	-0.3786	0.1444	0.0011	-0.3786	0.1444	0.0011
$\alpha = 0.1$						
50	-0.3516	0.1477	0.0241	-0.2490	0.0860	0.0240
200	-0.3772	0.1480	0.0057	-0.2750	0.0812	0.0056
1000	-0.3777	0.1438	0.0011	-0.2726	0.0755	0.0011
$\alpha = 0.25$						
50	-0.3490	0.1447	0.0230	-0.1162	0.0501	0.0366
200	-0.3783	0.1486	0.0055	-0.1647	0.0354	0.0083
1000	-0.3789	0.1447	0.0011	-0.1677	0.0297	0.0016
$\alpha = 0.5$						
50	-0.3509	0.1471	0.0240	-0.0981	0.0538	0.0442
200	-0.3767	0.1475	0.0056	-0.1088	0.0230	0.0112
1000	-0.3783	0.1442	0.0011	-0.1083	0.0139	0.0022
$\alpha = 0.75$						
50	-0.3519	0.1479	0.0241	-0.1732	0.0870	0.0570
200	-0.3777	0.1483	0.0056	-0.1676	0.0416	0.0135
1000	-0.3779	0.1439	0.0011	-0.1706	0.0317	0.0026
$\alpha = 0.9$						
50	-0.3462	0.1442	0.0244	-0.2264	0.1256	0.0744
200	-0.3781	0.1486	0.0056	-0.2239	0.0666	0.0165
1000	-0.3777	0.1438	0.0011	-0.2225	0.0527	0.0032

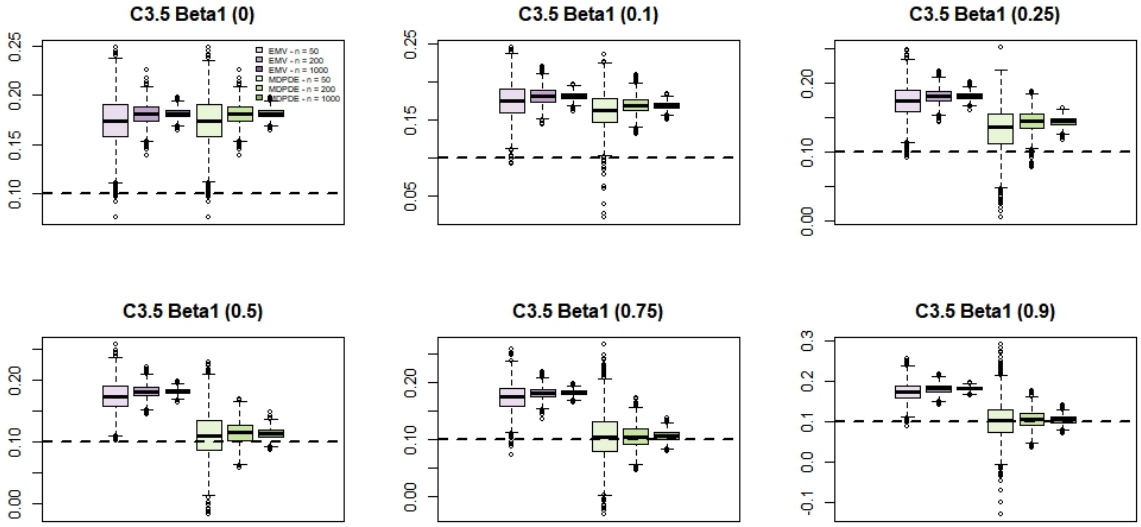


Figure 12: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 12: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	0.0739	0.0060	5e-04	0.0739	0.0060	0.0005
200	0.0815	0.0068	1e-04	0.0815	0.0068	0.0001
1000	0.0816	0.0067	0e+00	0.0816	0.0067	0.0000
$\alpha = 0.1$						
50	0.0740	0.0060	5e-04	0.0620	0.0044	0.0005
200	0.0812	0.0067	1e-04	0.0691	0.0049	0.0001
1000	0.0815	0.0067	0e+00	0.0687	0.0047	0.0000
$\alpha = 0.25$						
50	0.0740	0.0060	5e-04	0.0329	0.0022	0.0011
200	0.0814	0.0067	1e-04	0.0442	0.0022	0.0002
1000	0.0816	0.0067	0e+00	0.0447	0.0020	0.0000
$\alpha = 0.5$						
50	0.0739	0.0060	6e-04	0.0111	0.0014	0.0013
200	0.0812	0.0067	1e-04	0.0144	0.0005	0.0003
1000	0.0816	0.0067	0e+00	0.0142	0.0003	0.0001
$\alpha = 0.75$						
50	0.0742	0.0061	6e-04	0.0056	0.0016	0.0016
200	0.0814	0.0067	1e-04	0.0056	0.0004	0.0004
1000	0.0815	0.0067	0e+00	0.0060	0.0001	0.0001
$\alpha = 0.9$						
50	0.0733	0.0059	6e-04	0.0036	0.0020	0.0020
200	0.0812	0.0067	1e-04	0.0046	0.0005	0.0005
1000	0.0815	0.0067	0e+00	0.0045	0.0001	0.0001

For β_1, Figure 12 and Table 12 demonstrates that the MDPDE achieves superior overall performance.

In this 5% contamination scenario, the application of a robust method is well justified, as the MLE suffers in the trade-off against the MDPDE.

3.7 Scenario C3 (10% of contamination):

In this scenario, 10% of the observations simultaneously had their responses replaced by a value generated from a Gamma distribution with a mean 5 times the original value, and their respective values for the covariate x_{i1} generated from a continuous uniform distribution on the interval $(16, 17)$.

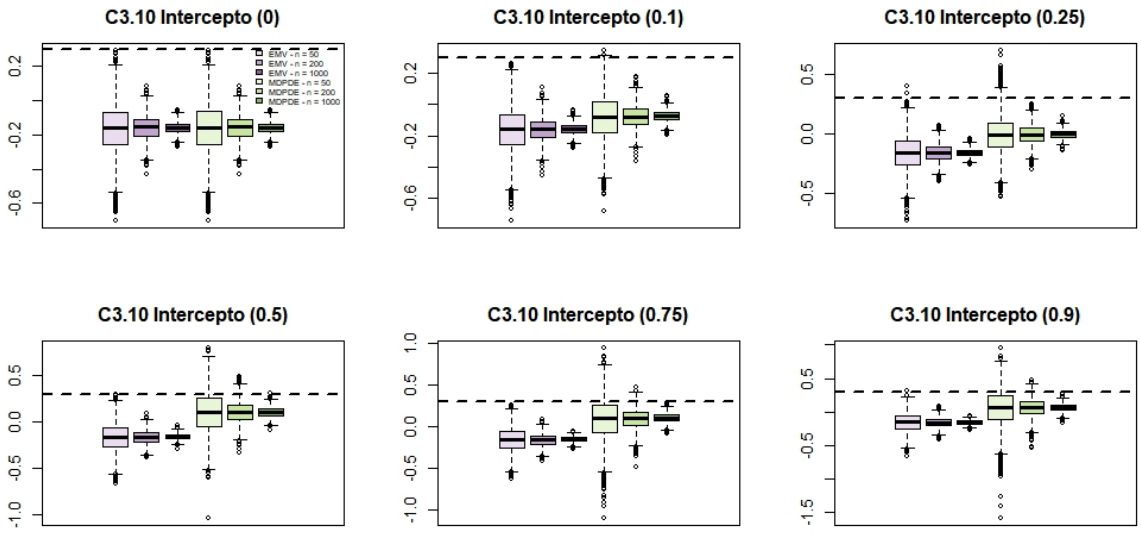


Figure 13: Distributions of intercept estimates for each sample size.

As expected, for $\alpha = 0$, the MLE and MDPDE share the exact same distribution of estimates. However, for α around 0.5, the robust method appears to perform better.

In Table 13, we observe that the MDPDE is consistently better than the MLE in terms of MSE.

Table 13: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Intercepto

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	-0.4627	0.2348	0.0207	-0.4627	0.2348	0.0207
200	-0.4582	0.2151	0.0051	-0.4582	0.2151	0.0051
1000	-0.4583	0.2111	0.0010	-0.4583	0.2111	0.0010
$\alpha = 0.1$						
50	-0.4626	0.2353	0.0213	-0.3811	0.1665	0.0212
200	-0.4598	0.2165	0.0050	-0.3768	0.1469	0.0050
1000	-0.4585	0.2113	0.0010	-0.3749	0.1416	0.0010
$\alpha = 0.25$						
50	-0.4628	0.2351	0.0210	-0.3080	0.1214	0.0265
200	-0.4605	0.2172	0.0051	-0.3070	0.1003	0.0060
1000	-0.4588	0.2116	0.0010	-0.3055	0.0945	0.0012
$\alpha = 0.5$						
50	-0.4653	0.2376	0.0211	-0.1942	0.0876	0.0499
200	-0.4619	0.2184	0.0051	-0.1953	0.0511	0.0130
1000	-0.4586	0.2113	0.0010	-0.1965	0.0413	0.0027
$\alpha = 0.75$						
50	-0.4616	0.2337	0.0206	-0.2134	0.1101	0.0645
200	-0.4612	0.2180	0.0053	-0.2073	0.0579	0.0149
1000	-0.4584	0.2112	0.0010	-0.1981	0.0422	0.0030
$\alpha = 0.9$						
50	-0.4616	0.2342	0.0211	-0.2466	0.1358	0.0750
200	-0.4593	0.2161	0.0051	-0.2397	0.0763	0.0188
1000	-0.4587	0.2114	0.0010	-0.2392	0.0605	0.0033

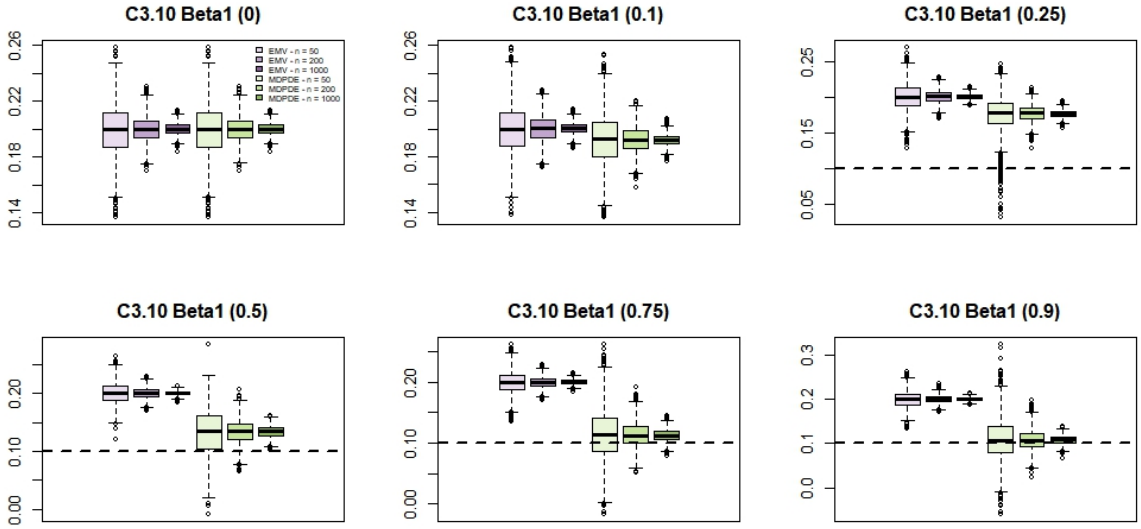


Figure 14: Distribuições das estimativas do beta_1 para cada tamanho de amostra.

Table 14: (Bias, MSE, and Variance) between MLE and MDPDE across sample sizes (n) and tuning parameters (α) - Beta1

	MLE			MDPDE		
n	Bias	MSE	Variance	Bias	MSE	Variance
$\alpha = 0$						
50	0.0994	0.0102	3e-04	0.0994	0.0102	0.0003
200	0.1000	0.0101	1e-04	0.1000	0.0101	0.0001
1000	0.1001	0.0100	0e+00	0.1001	0.0100	0.0000
$\alpha = 0.1$						
50	0.0999	0.0103	3e-04	0.0926	0.0089	0.0003
200	0.1002	0.0101	1e-04	0.0922	0.0086	0.0001
1000	0.1001	0.0100	0e+00	0.0920	0.0085	0.0000
$\alpha = 0.25$						
50	0.0996	0.0102	3e-04	0.0758	0.0063	0.0006
200	0.1003	0.0101	1e-04	0.0764	0.0060	0.0001
1000	0.1002	0.0101	0e+00	0.0762	0.0058	0.0000
$\alpha = 0.5$						
50	0.1001	0.0103	3e-04	0.0325	0.0026	0.0015
200	0.1003	0.0101	1e-04	0.0332	0.0015	0.0004
1000	0.1001	0.0100	0e+00	0.0338	0.0012	0.0001
$\alpha = 0.75$						
50	0.0999	0.0103	3e-04	0.0149	0.0020	0.0018
200	0.1001	0.0101	1e-04	0.0138	0.0006	0.0004
1000	0.1001	0.0100	0e+00	0.0124	0.0002	0.0001
$\alpha = 0.9$						
50	0.0995	0.0102	3e-04	0.0085	0.0021	0.0020
200	0.1000	0.0101	1e-04	0.0083	0.0006	0.0005
1000	0.1001	0.0100	0e+00	0.0082	0.0002	0.0001

For β_1, Figure 14 and Table 14 show that the MDPDE achieves superior performance.

In the 10% contamination scenario, the MLE suffers even more in the trade-off against the MDPDE, exhibiting larger MSE values for both parameters compared to the 5% contamination case (Tables 11 and 12). These results further justify the application of robust estimation techniques.

4 Conclusion

Overall, the comparative evaluation of the MLE and MDPDE highlights the crucial role of the tuning parameter α and the level of contamination. In the absence of contamination, both estimators coincide at $\alpha = 0$, with the MLE offering optimal efficiency. However, as contamination increases from 5% to 10%, the MLE suffers significant degradation in terms of MSE and dispersion for both the intercept and β_1. Conversely, setting $\alpha > 0$ (notably around 0.5 to 0.75) allows the MDPDE to maintain robust performance, achieving a lower MSE in larger samples and confirming that robust estimation is essential when data are subject to the types of contamination considered here.

5 References

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